Econ 8210: Problem Set #3

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1 Introduction

The main objective of this notebook is to study the incentives to price discriminate across locations under different levels of local competition. That is, what is the profitability associated with setting regional v local v uniform prices. This is in the spirit of DellaVigna and Gentzkow (2019) and Adams and Williams (2019).

Complete this problem set by answering the questions below. When you are finished, submit your completed work as pdf (or multiple pdfs) with naming convention **'PS1_[insert last name]'**. For example, 'PS1_Newberry.pdf'. Feel free to have one pdf for your write up and one for your code with the label "_code" and "_writeup", for example. Feel free to work together but turn in your own write up. You can use any programming language you wish

Part of the purpose of these problem sets it to help you learn how to write up an academic paper (especially an empirical one), so please include derivations of estimating equations, details of how you estimate the model, intuition about the answers, etc., in your write up.

2 Set up and Data Description

The data is based on a simulation of a retail industry. The simulation consists of 20 markets (t) and 3 firms (j). You can think of firm 1 as Walmart and firm 2 as Target. Note that market structure varies in terms of which firms are in which markets. This means that some markets will be more competitive than others. Specifically, Target enters fewer markets than Walmart. Each firm has 1 observed characteristic (x_{jt}, which is e.g., a measure of variety) and a scalar unobserved quality index (ξ_{jt}). I assume that the firms have, on average, the same variety, but the unobserved quality for Target is higher. The marginal costs are also higher for Target (on average). Retailers set prices competitively, but set them regionally (i.e., there are pricing zones that are made up of multiple markets). Therefore, prices are denoted p_{jz} where z indicates the pricing zone.

Consumer i's indirect utlity for retailer j in market t is give by:

$$u_{ijt} = \beta_0 + \gamma_i x_{jt} + \alpha_i p_{jz} + \xi_{jt} + \epsilon_{ijt}$$

where ϵ_{ijt} is assumed to be distributed according to an iid EV distribution. Consumer preferences vary by their income such that:

$$\gamma_i = \gamma_0 + \gamma_1 * y_i$$

and:

$$\alpha_i = \alpha_0 + \alpha_1 * y_i$$

where y_i is the consumer's income level. The utility can be written in the standard way as:

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \epsilon_{ijt}$$

The utlity of the outside option is:

$$u_{i0t} = \epsilon_{i0t}$$

where ϵ_{i0t} is distributed according to an iid EV distribution. Therefore choice probablity for consumer i and retailer $j \in J_t$, where J_t is the set of firms active in market t, is given by:

$$P_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{j' \in J_t} \exp(\delta_{j't} + \mu_{ij't})}$$

The income distribution varies by market and is given by F_t^y, which is a log normal distribution with mean μ and standard deviation σ. Therefore, the market share for retailer j in market t is given by:

$$s_{jt} = \int_{y_i \in t} P_{ijt} dF_t^y$$

Firm j's profit in market t is given by:

$$\pi_{jt} = M_t * s_{jt} * (p_{jz} - c_{jt})$$

and profit in zone z is:

$$\pi_{jz} = \sum_{j \in J_z} M_t * s_{jt} * (p_{jz} - c_{jt})$$

In the simulation, I set $\beta_0 = -1$, $\gamma_0 = 2$, $\gamma_1 = .02$, $\alpha_0 = -2$, $\alpha_1 = 0.05$.

1. Import the data called "full_data_zone.csv" on eLC. The data dictionary is on eLC in a file called "data_dictionary.csv". There you can learn what each variable is. Each observation is a market. I have also uploaded a "long" version of the data where each observation is a firm/market (called "full_data_zone_long.csv").

2. Create tables and/or graphs describing the interesting patterns in the data. For instance: what are the costs, level of competition, income, etc, for the markets that each firm is active in? Show evidence of zone pricing despite differences across markets (e.g., table 2 in Adams and Williams). Hint: look for a couple of particularly interesting zones and then create a table that shows the differences across markets within that zone.(ASIDE: I am intentionally leaving this somewhat open ended, because I want you to think hard about the best ways to explore the data.) Based on this information, which firms do you think benefits the most more or less coarse pricing?

3. Calculate the profit for each firm in each market. Assume that the market size is the same for all markets, such that $M_t = 10000 \ \forall \ t$. What are the characteristics of markets that are more profitable for each firm?

3 Estimating the Model

Now we will turn to estimating the model. Note that we are not estimating anything on the supply side because we observe the marginal cost.

1. Estimate the homogenous MNL model:
   (a) Without instrumenting for price. Are the parameters you get close to the true parameters? Which way are the biased and why?
   (b) Now estimate the model instrumenting for price. What is a good instrument to use (hint: think about the level of variation in prices)? Are the parameters you get close to the true parameters? Which way are the biased and why?

2. Estimate the hetergoenous (true) model with income interactions:
   (a) Instrumenting for price using the same instrument you used before. What is a good instrument for the income interactions? Are the parameters you get close to the true parameters? Which way are the biased and why?

3. Bonus: estimate the heterogenous model adding the micro moments. Are the parameters you get close to the true parameters? Which way are the biased and why?

4 Counterfactuals

Using your estimates from 2.2 (or 2.3) above, calculate profit of each market and overall profit for each retailer under 9 different pricing regimes (e.g., all different combinations of uniform, zone, and market-level pricing across the 2 firms).

1. Write out the profit function and first order condition for firm j, assuming uniform, zone, and market-based pricing, respectively?
2. Calculate and compare profit under each regime. Who benefits and who is hurt? What types of markets (i.e., market structures) are profitable for the different firms under each regime?
3. Calculate and compare consumer welfare under each regime. What types of markets are beneficial to consumer under each regime?